

Range-Proved Finger Convergence: Bounded Chord Finger Maintenance under Transport Admission and Browser Churn

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Abstract

Chord refreshes one finger per stabilization round, so a 160-slot table needs of the order of 160 rounds and a hinted entry is indistinguishable from a proved one. Rings runs Chord over browsers whose WebRTC handshakes take minutes or fail. We specify its finger maintenance as one pure transition step : $(S, \text{Event}) \rightarrow (S, \text{Actions})$ built on Chord's range lemma (a lookup at slot i answered at distance d proves every slot j with $2^j \leq d$) and four disciplines: per-slot evidence epochs, a single attempt owner `Idle` | `AwR` | `AwA` keyed by a fresh UUID, cyclic starvation-free selection with a deterministic retry floor, and single-use claims that bound each topology report to $K+1$ connections. We state the laws, prove range, starvation-freedom, and effect bounds, bound the per-node cost, and list the obligations discharged by the shipped model checker.

Table I
Chord finger maintenance versus this design

concern	Chord 2003 [2]	here
refresh unit	one slot per period	one range per lookup (Lem. 1)
local range	routed lookups	proved by stabilization (Lem. 2)
hinted entry	= proved entry	$\text{ver}_j = \perp$ until proved (§IV)
correlation	slot index	(i, id, e) (Law 5)
selection	random	cyclic cursor (Prop. 1)
report fan-out	unbounded	$\leq K+1$ (Prop. 2)

I. Introduction

Chord [1] routes in $O(\log N)$ hops because node n keeps finger $i = \text{succ}(n + 2^i)$. The journal revision [2] refreshes one random finger per period; an m -bit table converges in $\Theta(m)$ periods, and an entry copied from a neighbour is never distinguished from a proved one. Rings differs on two axes: members are browsers, so a routing entry is usable only after a WebRTC handshake that takes seconds to minutes and often fails; and $N \ll 2^{160}$, so a 160-slot table has only $O(\log N)$ distinct values. Table I summarizes what this design changes; the rest of the paper states each change as a law over step and proves the bound it buys.

II. Model

Definition 1 (Ring). Identifiers are $\mathbb{Z}/2^m$, $m = 160$; $\text{dist}(a, b) = (b - a) \bmod 2^m$. For a membership view V , $\text{succ}_V(x) = \arg \min_{s \in V} \text{dist}(x, s)$ and $\text{pred}_V(x) = \arg \max_{s \in V} \text{dist}(x, s)$. Node n keeps the K nearest clock-wise members ($K = 3$); $\text{head}(n)$ is the first of them other than n , or $\perp$.

Ryan J. Kung can be reached at ryankung@ieee.org. Manuscript created September 15, 2026. This document specifies the finger-table maintenance shipped in Rings 0.26.0 (rings-core, modules dht::finger::convergence and dht::topology).

Definition 2 (Finger slot). Slot $i \in [0, L)$, $L \leq m$, has target $t_i = n + 2^i$ and true value $\text{succ}_V(t_i)$. The table stores $\text{hint}_i \in V \cup \{\perp\}$; a value equal to n is stored as $\perp$.

Definition 3 (Hint join). For a learned peer $p \neq n$ with $d = \text{dist}(n, p)$:

$$\forall i : 2^i \leq d \wedge (\text{hint}_i = \perp \vee \text{dist}(n, \text{hint}_i) > d) \Rightarrow \text{hint}_i \leftarrow p.$$

Removing p clears every slot holding it and refills from the nearest surviving hint above. Neither operation produces proof.

Assumption 1 (Authenticated reports). Every report is signed by its origin and rejected unless the origin is the peer queried. Attribution is transport's; solicitation is what the tokens of §V and §VII establish.

Assumption 2 (Clocks). Finger deadlines are process-monotonic milliseconds; transport handshake generations are wall-clock. The two are never compared.

Assumption 3 (Effect boundary). step : $(S, \text{Event}) \rightarrow (S, \text{Actions})$ is pure. Clock readings and fresh UUIDs [3] arrive as event fields; sends, locks, and randomness happen only in the interpreter of the returned actions. Every law below is a property of step alone.

III. The Range Lemma

Lemma 1 (Chord range lemma [1]). Let $s = \text{succ}_V(t_i)$ and $d = \text{dist}(n, s)$. Then

$$\forall j \geq i : 2^j \leq d \Rightarrow \text{succ}_V(t_j) = s.$$

Proof. $[t_i, s) \cap V = \emptyset$ by definition of succ_V . For such j , $2^i \leq 2^j \leq d$ gives $t_j \in [t_i, s]$, so the first member at or after t_j is s . $\square$

This is Chord's `init_finger_table` test `finger[i].node  $\geq$  finger[i+1].start` [1, Fig. 6], used here as a proof rule.

Algorithm 1 $\text{Admit}(\pi)$: commit a range proof slot-wise

Require: $\pi = (i, \text{id}, e_\pi, s, \text{end})$, evidence $(e, (\text{chg}_j, \text{ver}_j))$

- 1: $A(\pi) \leftarrow \{j \in [i, \text{end}] : \text{chg}_j \leq e_\pi\}$
 - 2: if $A(\pi) = \emptyset$ then
 - 3: return **Stale**
 - 4: for all $j \in A(\pi)$ do
 - 5: $\text{hint}_j \leftarrow (s = n ? \perp : s)$; $\text{ver}_j \leftarrow \top$
 - 6: return **Applied**(end)
-

Definition 4 (Range proof). A lookup for slot i answered by s yields $\pi = (i, \text{id}, e_\pi, s, \text{end})$, with id the lookup UUID, e_π the epoch at issue (§IV), and

$$\text{end}(i, s) = \begin{cases} L-1 & s = n, \\ \min(L-1, \lfloor \log_2 \text{dist}(n, s) \rfloor) & 2^i \leq \text{dist}(n, s), \\ \text{Invalid} & \text{otherwise.} \end{cases}$$

π covers slots $i.. \text{end}$; $s = n$ means the arc from t_i wrapped to n without meeting a member.

Corollary 1 (Lookups per pass). If the true values partition the slots into R maximal runs of equal successor, a full pass needs at most R routed lookups, and $R \leq \min(L, |V|)$; under uniform identifiers $\mathbb{E}[R] = O(\log N)$.

Lemma 2 (Local range). Let $h = \text{head}(n)$ and let n hold a stabilization report from h , answering n 's own query, stating $\text{pred}(h) = n$. Then $\forall j : 2^j \leq \text{dist}(n, h) \Rightarrow \text{succ}_V(t_j) = h$.

Proof. $\text{pred}(h) = n$ gives $(n, h) \cap V = \emptyset$; $t_j \in (n, h]$ iff $2^j \leq \text{dist}(n, h)$. $\square$

A routed lookup for $t_j \in (n, h]$ would travel the whole ring ($2O(\log N)$ hops) for a fact the stabilization exchange already carries, so the local range never sends one. The report must echo the current stabilization token and h must be the head both when queried and after the report is applied.

IV. Evidence Epochs

Definition 5 (Evidence). The evidence state is $(e, (\text{chg}_j, \text{ver}_j)_{j < L})$, $e \in \mathbb{N}$ the epoch, chg_j the epoch at which hint_j last changed, ver_j a bit. Initially $e = 0$, $\text{chg}_j = 0$, $\text{ver}_j = \perp$.

Law 1 (Hint change). For a topology transition $H \rightarrow H'$ with $C = \{j : H_j \neq H'_j\} \neq \emptyset$: $e \leftarrow e + 1$ and $\forall j \in C : (\text{chg}_j, \text{ver}_j) \leftarrow (e, \perp)$. On overflow of e : $\forall j : \text{ver}_j \leftarrow \perp$ and the attempt is retired.

Law 2 (Admission). Proof π may write slot $j \in [\pi.i, \pi.\text{end}]$ iff $\text{chg}_j \leq e_\pi$. It is accepted iff some slot admits it (Algorithm 1); admitted slots are written, the rest are skipped, none is rolled back. Acceptance and commit are the same per-slot predicate, so an accepted proof always writes.

Law 3 (Discontinuity). Losing the last successor sets $e \leftarrow e + 1$, $\forall j : \text{ver}_j \leftarrow \perp$, $\alpha \leftarrow \text{Idle}$. Ranges stay unverified

but **Dormant** until a successor is admitted again; isolation is not proof of emptiness.

Law 4 (Local confirmation). Applying Lemma 2 sets $\text{ver}_j \leftarrow \top$ for $j \leq \lfloor \log_2 \text{dist}(n, h) \rfloor$ with no lookup. If α 's slot lies in that range, $\text{Progress}(\lfloor \log_2 \text{dist}(n, h) \rfloor)$ (Law 7): superseded, not failed.

V. Attempt Ownership

Definition 6 (Attempt). At most one attempt α exists:

$$\alpha \in \text{Idle} \mid \text{AwR}(\rho, e_\rho, \tau_r) \mid \text{AwA}(\pi, \tau_a),$$

$\rho = (i, \text{id})$ the request, e_ρ the epoch at issue, τ_r the report deadline, π a validated proof, τ_a the admission lease. Two owners are not representable.

Law 5 (Consumption). A report (ρ', s') consumes the attempt iff

$$[\text{AwR}(\rho, \cdot, \cdot) \wedge \rho' = \rho] \vee [\text{AwA}(\pi, \cdot) \wedge \rho' = \pi.\rho \wedge s' = \pi.s].$$

Any other report is **Stale**: no state change, no failure.

Law 6 (Deadlines).

$$\begin{aligned} \tau_r &= \text{issued} + T_r, & \tau_a &= \text{deferred} + T_a, \\ T_r &= 10 \text{ s}, & T_h &= 180 \text{ s}, & T_a &= T_h + 30 \text{ s}. \end{aligned}$$

T_h is the transport's handshake-generation lifetime.

A proof is deferred to **AwA** the moment a timely report names an unconnected peer, before any handshake is awaited, so a cancelled handler cannot strand it; the generation, once reserved, releases the proof by its own admission or expiry. $T_a > T_h$ makes the lease a backstop that cannot expire first and discard a proof whose handshake later succeeds; the converse (committing a released proof) is excluded by Law 5.

Law 7 (Outcomes). A consuming report is validated in the order token, deadline, geometry (Definition 4), epoch (Law 2); Algorithm 2 gives the transitions in terms of the failure streak k and retry floor ν of Definition 9 and three named transitions on (α, c, k, ν) :

$$\text{Fail}(i) : (\text{Idle}, \text{run}(i) + 1, k + 1, t + f(k + 1))$$

$$\text{Retire}(\text{end}) : (\text{Idle}, \text{end} + 1, k + 1, t + f(k + 1))$$

$$\text{Progress}(\text{end}) : (\text{Idle}, \text{end} + 1, 0, \perp).$$

A hint change at the attempt's lower slot (Law 1) destroys the premise of its proof and retires the owner without raising k : churn is new evidence, not a network failure [4].

VI. Selection and Retry

Definition 7 (Routable region). With $h = \text{head}(n)$, the first routable slot is $r = \lfloor \log_2 \text{dist}(n, h) \rfloor + 1$ ($r = 0$ if $h = \perp$). Slots below r are the local range and are never selected.

Algorithm 2 Report($\rho' = (i, \text{id}', s', t, a)$; a is the transport's verdict on s')

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1: if  $\neg \text{consumes}(\alpha, \rho', s')$  then
2:   return Stale ▷ Law 5
3: if  $t \geq \tau$  then
4:   Fail( $i$ ); return Expired
5: if  $\alpha = \text{AwA}(\pi, \cdot)$  then
6:    $\pi' \leftarrow \pi$  ▷ already validated
7: else
8:   end  $\leftarrow \text{end}(i, s')$ 
9:   if end = Invalid then
10:    Fail( $i$ ); return Invalid
11:    $\pi' \leftarrow (i, \text{id}', e_\rho, s', \text{end})$ 
12:   if  $A(\pi') = \emptyset$  then
13:     $\alpha \leftarrow \text{Idle}$ ; return Stale ▷ churn, not failure
14: if  $a = \text{Connected}$  then
15:   Admit( $\pi'$ ); Progress(end)
16: else if  $a = \text{Pending}$  then
17:    $\alpha \leftarrow \text{AwA}(\pi', t + T_a)$  ▷ deferred
18: else
19:   Retire(end) ▷ unroutable

```

Definition 8 (Cursor and runs). The state carries $c \in \mathbb{N}$, initially 0. With $A = \{j \geq \max(c, r) : \neg \text{ver}_j\}$ and $B = \{j \in [r, \max(c, r)) : \neg \text{ver}_j\}$,

$$\text{sel}(c) = \begin{cases} \min A & A \neq \emptyset \\ \min B & B \neq \emptyset \\ \perp & \text{otherwise,} \end{cases}$$

$$\text{run}(i) = \max\{j \geq i : \forall l \in [i, j], \text{hint}_l = \text{hint}_i\}.$$

After an applied proof or a retired validated proof, $c \leftarrow \text{end} + 1$; after Fail(i), $c \leftarrow \text{run}(i) + 1$, because slots sharing a hint expect the same successor.

Definition 9 (Retry floor). Retry state (ℓ, k, ν) : last issue time, failure streak $k \in [0, 255]$ (saturating), earliest retry. Issue at t is permitted iff

$$t - \ell \geq T_{\min} = 1 \text{ s} \wedge t \geq \nu, \quad f(k) = \min(2^{\min(k, 6)} \text{ s}, 60 \text{ s}).$$

Progress sets $k \leftarrow 0$, $\nu \leftarrow \perp$ and keeps ℓ , so progress never bypasses $T_{\min}$.

Law 8 (Issue). A tick is Algorithm 3. At most one lookup is emitted per tick; the expiry tick emits none so the scheduler observes the new floor.

Proposition 1 (Starvation freedom). Let U be the routable slots that stay unverified across a sequence of attempts. Between two selections of $u \in U$, every other slot of U is selected at least once.

Proof. Each attempt at j moves c strictly past j ($\text{end} + 1$ or $\text{run}(j) + 1$, both $> j$), and sel returns the first unverified routable slot cyclically from c . Members of U are unverified at every selection, so the selected sequence visits U in cyclic index order, once per cycle. $\square$

Algorithm 3 Tick(t, id): retire an expired attempt, else issue

```

1: if  $\alpha \neq \text{Idle} \wedge t \geq \tau$  then
2:   Fail( $\alpha.i$ ); return  $\emptyset$ 
3: if  $\alpha \neq \text{Idle} \vee \forall j \text{ver}_j \vee t - \ell < T_{\min} \vee t < \nu$  then
4:   return  $\emptyset$ 
5:  $i \leftarrow \text{sel}(c)$ ; if  $i = \perp$  then return  $\emptyset$ 
6:  $\alpha \leftarrow \text{AwR}((i, \text{id}), e, t + T_r)$ ;  $\ell \leftarrow t$ 
7: if  $t_i \in (n, \text{head}(n)]$  then
8:   apply Law 4; return  $\emptyset$ 
9: else
10:  return  $\{\text{Lookup}(i, \text{id})\}$ 

```

The first shipped design selected $\min\{j \geq r : \neg \text{ver}_j\}$; one slot whose successor never admits (the NAT case) then held the single attempt forever.

Law 9 (Revalidation gate). A periodic revalidation sets $\text{ver}_j \leftarrow \perp$ for $j \in [c', \text{run}(c')]$, $c' = c \bmod L$, iff

$$\alpha = \text{Idle} \wedge (\neg \exists j \geq r : \neg \text{ver}_j \vee k > 0).$$

Pending routable work defers revalidation only while succeeding. Without the $k > 0$ disjunct one permanently failing slot would freeze the revalidation of every proved range; with it, Proposition 1 serves the reopened range before the failing slot is retried.

VII. Report Claims

Two other report-driven flows may open connections: the stabilization report (the head's successor list and predecessor) and the successor-list synchronization report. Each is correlated by a two-phase token.

Definition 10 (Tokens). The stabilization token is at most one $\text{tok} = (h, \text{id}, \phi)$, $\phi \in \{\text{Req}, \text{Proc}\}$. Sync tokens form a map $p \mapsto (\text{id}, \phi)$ with at most one entry per current successor.

Law 10 (Stabilization round). The round is Algorithm 4. Claim is one transition (predicate and phase move), so two deliveries of one report cannot both claim. Candidates are ordered by transport quality. The release guard is constructed only by a successful claim; a head change clears tok unconditionally.

Law 11 (Sync round). Begin registers (id, Req) for p iff p is a current successor and p 's entry is not **Proc**, superseding an unanswered entry. Claim and per-candidate re-check follow Algorithm 4 with budget K . Every successor-list commit prunes entries whose reporter left the list and keeps the rest, so a plan keeps its budget while its reporter stays, including when its own admissions extend the list.

Proposition 2 (Effect bound). One stabilization report causes at most $K + 1$ connection attempts and one topology commit; one sync report at most K . After a token is cleared, at most the one connection already permitted completes.

Algorithm 4 Stabilization round for reporter h' , token id'

```

1: procedure Begin(id)
2:   if head( $n$ ) =  $\perp$  then
3:     tok  $\leftarrow \perp$ ; return  $\emptyset$ 
4:   if tok. $\phi$  = Proc then
5:     return  $\emptyset$   $\triangleright$  never revoke a live claim
6:   tok  $\leftarrow$  (head( $n$ ), id, Req); return {Query(tok)}
7: procedure Handle( $h'$ , id', pred', succs')
8:   if tok  $\neq$  ( $h'$ , id', Req) then
9:     return  $\triangleright$  unclaimed: dropped
10:  tok. $\phi$   $\leftarrow$  Proc; guard  $\leftarrow$  Release(id') on every exit
11:   $C \leftarrow$  first  $K+1$  distinct of (pred', succs')  $\setminus \{n\}$ 
12:  for all  $x \in C$  do
13:    if tok  $\neq$  ( $h'$ , id', Proc) then
14:      return
15:    await Connect( $x$ )
16:  if tok = ( $h'$ , id', Proc) then
17:    merge succs'; emit improvement query, notify
18:    if head( $n$ ) =  $h'$   $\wedge$  pred' =  $n$  then
19:      confirm  $[0, \lfloor \log_2 \text{dist}(n, h') \rfloor]$   $\triangleright$  Lem. 2
20:    tok  $\leftarrow \perp$ 
21: procedure Release(id')
22:   if tok.id = id' then
23:     tok  $\leftarrow \perp$ 

```

Proof. C is bounded before any per-candidate effect; the cursor over C advances only on a successful re-check, and a cleared token fails every later re-check. Apply requires Proc and clears the token, so it happens at most once. $\square$

VIII. Scheduler

The scheduler is the only reader of a wall clock and the only source of randomness, drawn from a replayable state seeded by the node identifier and a per-lifecycle UUID.

Law 12 (Wake time). The finger phase at poll time t is one of Dormant (no head), Converged (no unverified routable slot, no attempt), Runnable, AwR, AwA, with streak k and remaining lease Δ . With $f(0) = 1$ s and $\mathcal{U}[0, x]$ a uniform jitter draw,

$$\text{wake}(t) = \begin{cases} t + \Delta & \text{(a) AwR } \vee \text{ AwA,} \\ t + 1 \text{ s} + \mathcal{U}[0, 10 \text{ s}] & \text{(b) activation,} \\ t + f(k) + \mathcal{U}[0, f(k)] & \text{(c) re-entry,} \\ \text{unchanged} & \text{(d) otherwise,} \end{cases}$$

where (b) is Dormant $\rightarrow$ Runnable with $k = 0$ (first successor admitted), (c) any other entry into Runnable or a change of k , and (d) Runnable unchanged, so polling cannot postpone work. A finger turn completing at t' schedules $t' + f(k) + \mathcal{U}[0, f(k)]$ from t' , never from the missed deadline.

Law 13 (Suspension). If the gap since the last poll and the lateness of a runnable finger deadline are both ≥ 10 s, the deadline is redrawn as $t + 1 \text{ s} + \mathcal{U}[0, 10 \text{ s}]$ and priority deferrals reset. A retained proof stays dormant until τ_a .

The lifecycle UUID belongs to the ring, not the listener: restarting a listener replays the phase; a new ring draws a new one. A finger turn yields to a simultaneously due topology or storage phase at most twice.

IX. State Relation

The following hold in every reachable state of step.

Invariant 1 (Proof freshness). $\text{ver}_j = \top \Rightarrow$ some π with $j \in [\pi.i, \pi.\text{end}]$ and $\pi.e \geq \text{chg}_j$ was applied, or Law 4 confirmed j , after the last change of hint_j .

Invariant 2 (Single owner). α is exactly one of Idle, AwR, AwA; request identifiers issued by one process are distinct; consecutive issues are $\geq T_{\min}$ apart.

Invariant 3 (Exact consumption). Every report-caused state change is caused by a report satisfying Law 5; a Stale report changes neither state nor k .

Invariant 4 (Failure accounting). k increases only by Fail or Retire, returns to 0 only by Progress or Law 4, and ν is set only together with an increase of k .

Invariant 5 (Cyclic service). Selected slots form a cyclic sequence over the unverified routable slots (Proposition 1).

Invariant 6 (Token discipline). A stabilization or sync report is applied, or opens a connection, only while its exact (reporter, id) is Proc; a token is Proc only while a live handler holds its claim; a head change or the reporter leaving the successor list clears it.

Invariant 7 (Head law). SuccessorHeadChanged(h) is the last action of a transition iff the head moved to h ; the stabilization token is cleared in exactly those transitions.

X. Cost

Proposition 3 (Per-node cost). Per full pass, routed finger lookups number at most R (Corollary 1) plus the ranges reopened by revalidation during the pass, each $O(\log N)$ hops and one report; the local range costs zero. Emissions are $\geq T_{\min}$ apart and, after k failures, $\geq f(k)$ apart. Per stabilization period: one head query and $\leq K+1$ connection attempts; per sync report $\leq K$.

A slot whose successor never admits costs one attempt per rotation (a lookup, a deferral, and up to T_h of handshake); by Proposition 1 the rest of the table still converges once per rotation. Fleet startup is $O(N \log N)$ lookups, each node's spread over its own $[1, 11]$ s activation window; this is a per-node bound, not a global rate limit on a seed.

XI. Security

The adversary controls any number of identifiers, delivery order and timing, and every node other than n ; it cannot forge signatures (Assumption 1). Sybil count is out of scope. Table II lists what each mechanism bounds.

Table II
Threats and the law that bounds each

threat	bound
unsolicited report	finger: must consume the attempt (Law 5); topology: Invariant 6. Only the queried peer matches the origin, and it may answer its one query once.
replay, reordering	a delayed report carries an unowned UUID (Stale); one issued under $e < e'$ after a hint change is rejected slot-wise (Law 2); a restart rebuilds table and attempt empty, so no pre-restart report matches.
wrong answer	$2^i > \text{dist}(n, s) \Rightarrow \text{Invalid}$, charged as failure; otherwise a hint verified relative to that peer's view. A lying successor affects routing efficiency, never ring safety [5], and the range is revalidated.
amplification	one report $\Rightarrow \leq K+1$ (or K) connections, bounded before any effect; one finger report $\Rightarrow \leq 1$ handshake; one attempt outstanding per node; a refusing peer costs one attempt per rotation (Prop. 1).
clock manipulation	deadlines are process-monotonic (Assumption 2); wall-clock changes neither extend nor expire an attempt.

XII. Verification

The implementation executes step directly; the obligations of Table III are discharged against it, not against a shadow model.

XIII. Related Work

Chord's conference paper [1] contains both the range lemma and the hint model; its journal revision [2] replaced both with randomized refresh. Zave [5], [6] proved ring correctness rests on the successor list alone and fingers are irrelevant to safety, the licence under which this design treats every finger as revocable evidence. Bamboo [4] set the rule that maintenance backs off on timeouts, never on new evidence (last branch of Algorithm 2). Kademlia's k -buckets [7] are the nearest analogue of a proved range, without epoch or owner. The evidence epoch is a length-one version vector per slot [8].

XIV. Conclusion

Finger convergence in Rings is Chord's range lemma under four disciplines the original did not need: evidence epochs, one correlated attempt owner, cyclic selection with a deterministic retry floor, and single-use claims on connection-opening reports. Each is a law over one pure transition, checked against that transition, and together they bound what one node sends, what one report can cause, and how long one unreachable peer can delay the rest of the table.

Table III
Obligations discharged by the shipped tests

obligation	driver and checked property
oracle convergence	sparse, dense, wrapped, and mutation-matrix rings driven with oracle answers reach succ_V on every slot with $\leq R$ routed lookups
resource laws	explorer over {advance, advance to deadline, progress, invalid, late, cancel, defer, admit deferred, lose, duplicate, topology change, restart} to depth 9 checks Invariants 2–4, deadline cleanup, and backoff growth; restart = empty table rejoining the previous head
claim discipline	explorer over {begin, claim, reserve, execute, superseded candidate, cancel, complete, superseded proof, unclaimed proof, move head} checks Proposition 2, that a report against a Req token changes nothing, and that begin never revokes Proc
starvation, gate	a cancelled attempt at the first remote range is followed by the next range; selection wraps to the failing range only after every other is proved; revalidation reopens a range while the only pending work is failing
transport cascade	a 3-node dummy-transport ring drains the missing-finger cascade within 20 control submissions; a 5-node ring with all targets local emits 0 finger submissions
schedule	over 200 deterministic nodes: activation in [1, 11]s, saturated retry in [60, 120]s, ≤ 30 nodes per one-second bucket, the lifecycle UUID alone changes the deadline, and re-entry from AwR or Converged draws [1, 2]s

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